

BACKGROUND PLACEHOLDER
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Quality Assurance (QA) Audit Report

Requirements Engineering Process

Abyss Navigator Suite Project

[QA AUDIT ORGANISATION NAME]

March 15, 2026

QA Team Member	Department	Role	Employee ID
Zong Han	QA Audit Division	QA Auditor	QA-001

Table 1: QA Audit Personnel

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1 Executive Summary

This report presents the results of a Quality Assurance (QA) audit of the Requirements Engineering (RE(-)) process used to produce the Software Requirements Specification (SRS) for the **Abyss Navigator Suite** project.

The purpose of this QA audit is to determine whether the vendor's conduct of Analysing, Specifying, and Validating followed disciplined and sufficiently documented Requirements Engineering practices. In accordance with the Validating method, QA targets the **process that produces the SRS**, rather than the SRS work product itself.

The audit focused on process non-compliance risks such as incomplete process evidence, weak traceability discipline, insufficient documentation of review activities, and uneven coverage of method-specific artefacts. While the RE(-) process shows a generally structured and credible approach, several areas require strengthening to improve auditability, process transparency, and confidence in method compliance.

Overall, the QA audit concludes that the RE(-) process is **conditionally acceptable**, subject to rectification of the identified process non-compliance issues before future downstream governance and project-control activities rely on the process record as complete and robust.

2 Introduction

2.1 Purpose

This Quality Assurance (QA) Audit Report evaluates the Requirements Engineering (RE(-)) process used in producing the Software Requirements Specification (SRS) for the **Abyss Navigator Suite** project.

While Quality Control (QC) focuses on detecting defects within the SRS work product, Quality Assurance focuses on evaluating whether the **process used to produce the requirements followed appropriate Requirements Engineering practices**.

The purpose of this QA audit is therefore to determine whether the RE(-) process applied during the project followed disciplined and systematic procedures.

2.2 Document Conventions

This document is an audit report rather than a process manual. It records detected process non-compliance issues, explains their implications, and recommends rectification actions.

Error detection instances use the format QA-ERR-XXX.

Audit findings use the format QA-FND-XXX.

Rectification recommendations use the format QA-REC-XXX.

2.3 Intended Audience and Reading Suggestions

This report is intended for Customer C-staff, project management, the requirements engineering team, and other stakeholders responsible for process governance in Requirements Engineering.

Readers should begin with the Executive Summary and Sections 1–3, then proceed to the QA Checklist, Error Detection, detailed Findings, and Recommended Process Rectifications.

2.4 Scope of QA Audit

The QA audit focuses on evaluating the process used during the following RE(-) activities:

- requirements analysis
- requirements specification
- requirements validation
- requirements traceability management
- requirements change-mitigation setup
- process evidence and governance records

3 Audit Basis

3.1 Nature of QA

Quality Assurance audits the **process used to produce requirements**, rather than the SRS document itself.

The QA auditor therefore evaluates whether the vendor team followed structured Requirements Engineering practices during Analysing, Specifying, and Validating.

3.2 Basis of Evaluation

The QA audit was conducted on the basis that the vendor should have followed process expectations associated with:

- regulated-revision during Analysing
- documented incorporation of development provisions during Specifying
- planned and report-based auditing during Validating
- maintenance of supporting plans, reports, checklists, meeting records, and artefacts throughout RE(-)

3.3 Kinds of Errors Pursued

The QA audit investigates whether the Requirements Engineering process suffered from failure to strictly comply with established RE(-) practices.

Potential causes examined include:

- insufficient or invalid process evidence
- insufficient resources allocated to requirements activities
- inadequate management tools supporting requirements work
- insufficient training or guidance in applying RE(-) techniques
- incomplete linkage between process tasks and artefacts

3.4 Severity Classification

Findings are categorised according to severity:

- **High** – process non-compliance that materially weakens confidence in whether RE(-) methods were properly conducted
- **Medium** – process weakness that reduces auditability, process clarity, or evidence quality
- **Low** – minor documentary or structural weakness that does not materially undermine overall process conduct

3.5 Records Reviewed

The following project records were examined during the QA audit:

- Project Specification (PS)
- Software Requirements Specification (SRS)
- requirements modelling artefacts
- requirement traceability records
- meeting documentation and review records
- project planning and documentation materials
- supporting notes relevant to Analysing, Specifying, and Validating conduct

4 QA Plan

4.1 Target Process Description

The target of this QA audit is the vendor's **Requirements Engineering process** used to transform the Project Specification into a validated SRS through Analysing, Specifying, and Validating.

4.2 Context and Goals

The RE(-) process is expected to convert customer-originated requirements content into a vendor-ready and validated SRS that can be relied upon by downstream development activities. Since the quality of the final SRS depends partly on the integrity of the process that produced it, QA is conducted to assess whether that process was sufficiently disciplined and auditable.

The goals of this QA audit are:

- to determine whether the RE(-) process followed systematic and sufficiently documented practices
- to determine whether sufficient evidence exists to support key requirements development activities
- to identify process weaknesses that may affect confidence in the quality and integrity of the resulting SRS

4.3 Access and Audit Conditions

The QA auditor reviewed documentation and artefacts produced during the requirements engineering stages. Access was limited to materials necessary for evaluating the RE(-) process and its supporting records.

4.4 Work Breakdown

Audit Role	Assigned Auditor	Primary Responsibility
QA Auditor	Zong Han	Review process records, evaluate method compliance, identify process non-compliance instances, and prepare the final QA report for Customer C-staff.

Table 2: QA audit work allocation

4.5 Indicative Timeline

Stage	Activity	Indicative Duration
Stage 1	Review process documentation and supporting artefacts for Analysing, Specifying, and Validating	2 hours
Stage 2	Map observed records against expected RE(-) method conduct	1 hour
Stage 3	Identify non-compliance instances and analyse severity	1 hour
Stage 4	Finalise QA report and sign-off submission to Customer C-staff	1 hour

Table 3: Indicative QA audit timeline

4.6 Reporting

The QA auditor documents findings and recommendations in this report. The auditor does not directly modify the audited work products or process records.

5 QA Checklist

The following checklist was used to inspect whether the RE(-) process showed sufficient compliance, evidence, and process discipline.

Item	Checklist Question	Inspection Result
Q1	Is there sufficient documented evidence that Analysing was carried out as regulated-revision rather than ad hoc rewriting?	Partially satisfied
Q2	Is there evidence that Specifying incorporated development provisions such as quality model, iRTM, and change mitigation?	Mostly satisfied
Q3	Is there evidence that Validating was conducted as an audit-oriented activity rather than informal review only?	Partially satisfied
Q4	Are process records such as plans, notes, and artefacts sufficiently linked to the three RE(-) methods?	Partially satisfied
Q5	Are traceability records and supporting models sufficiently governed as process artefacts?	Mostly satisfied
Q6	Are assumptions, constraints, and method decisions explicitly recorded where needed?	Partially satisfied
Q7	Is the process record sufficiently clear for an independent auditor to understand what was done and how?	Partially satisfied
Q8	Are supporting addenda and referenced process artefacts appropriately linked to the base report?	Mostly satisfied

Table 4: QA process checklist

6 QA Error Detection

The QA audit identified instances of **process non-compliance** in the vendor's conduct of Requirements Engineering. Each instance is recorded with its location and characteristic.

Error ID	Error Type	Location	Characteristic of Error
QA-ERR-001	Process compliance non-	Analysing process evidence	Process records do not always make the regulated-revision trail fully explicit from issue identification to corrected requirement outcome.
QA-ERR-002	Process compliance non-	Validation / review records	Documentation of formal stakeholder validation or clarification activity could be stronger in the retained process record.
QA-ERR-003	Process compliance non-	Traceability governance	The process record shows traceability artefacts, but the governance linkage between source decisions and downstream traceability maintenance can be clearer.
QA-ERR-004	Process compliance non-	Method documentation	Some assumptions, constraints, and supporting method decisions are present in artefacts but not always captured as explicit process evidence.

Table 5: Detected QA error instances

7 QA Findings

7.1 Finding Summary

Finding ID	Type	Severity	Short Description
QA-FND-001	Process evidence weakness	Medium	The regulated-revision trail for some analysed content is not always explicit enough for independent reconstruction of the process path.
QA-FND-002	Review / validation evidence weakness	Medium	Stronger retained evidence of stakeholder validation and clarification activity would improve process confidence.
QA-FND-003	Traceability governance weakness	Medium	Traceability is present, but the supporting process evidence for how it was governed can be made more explicit.
QA-FND-004	Documentation discipline weakness	Low	Some assumptions, constraints, and method decisions would benefit from more explicit process-level recording.

Table 6: Summary of QA findings

7.2 Detailed Findings

QA-FND-001: Regulated-revision trail not always explicit

Related error detection: QA-ERR-001

Type: Process evidence weakness

Severity: Medium

Context. Analysing in RE(-) is expected to function as regulated-revision of the Project Specification into ready-content for the SRS.

Result. The available process records support that analysis occurred, but some parts of the process trail are not always explicit enough for an independent reviewer to reconstruct the sequence from identified issue to final revised requirement state.

Analysis. This is a process non-compliance concern because the method may have been conducted substantively, yet the retained evidence does not always show the full disciplined pathway with sufficient visibility.

Message. Strengthen the retained record of regulated-revision activity by documenting issue identification, clarification input, interpretation outcome, and categorisation result more explicitly.

QA-FND-002: Stakeholder validation evidence can be stronger

Related error detection: QA-ERR-002

Type: Review / validation evidence weakness

Severity: Medium

Context. Requirements clarification and validation rely on cooperation between customer and vendor, especially where uncertainty or incongruity exists.

Result. The process artefacts suggest that validation and clarification occurred, but the formal retained evidence of those activities could be stronger and more consistently visible.

Analysis. This reduces confidence in strict process compliance because an independent auditor should be able to verify that required clarification and confirmation activities were actually carried out.

Message. Retain clearer meeting notes, review decisions, or other formal evidence showing how clarification and validation outcomes were achieved.

QA-FND-003: Traceability governance process needs clearer linkage

Related error detection: QA-ERR-003

Type: Traceability governance weakness

Severity: Medium

Context. Traceability is expected to operate as a governance mechanism linking requirements, artefacts, and downstream development records.

Result. Traceability artefacts exist, but the process record can more clearly show how traceability decisions, mappings, and maintenance responsibilities were governed.

Analysis. This is not a claim that traceability is absent. Rather, it is a process non-compliance concern that the supporting governance evidence is not as explicit as it could be.

Message. Add clearer process records showing how traceability mappings were established, checked, and maintained.

QA-FND-004: Some method decisions are insufficiently explicit as process evidence

Related error detection: QA-ERR-004

Type: Documentation discipline weakness

Severity: Low

Context. RE(-) process discipline benefits when assumptions, constraints, and method decisions are explicitly recorded rather than only inferable from artefacts.

Result. Some of these items appear to be represented in project materials, but not always in a form that makes the process trail explicit.

Analysis. This weakens process transparency slightly, although it does not on its own materially invalidate the overall RE(-) process.

Message. Record key assumptions, constraints, and process decisions more explicitly in retained process notes and supporting artefacts.

8 Recommended Process Rectifications

8.1 Priority Rectification Actions

Based on the process non-compliance issues identified during the QA audit, the following rectification actions are recommended in priority order:

1. strengthen the retained regulated-revision trail for analysed requirements content
2. improve retained evidence of clarification, review, and validation activities
3. make traceability governance decisions and maintenance responsibilities more explicit
4. document assumptions, constraints, and method decisions more systematically

8.2 Recommended Ownership of Rectifications

Recommendation ID	Owner	Action
QA-REC-001	Requirements Engineering Lead	Strengthen process evidence linking identified issues, clarification outcomes, and revised requirements.
QA-REC-002	Project Leader / Requirements Governance	Ensure validation and clarification sessions leave a stronger retained audit trail.
QA-REC-003	Traceability Custodian / Reviewer	Record clearer governance of traceability mappings, reviews, and maintenance actions.
QA-REC-004	Process Documentation Owner	Standardise recording of assumptions, constraints, and method decisions across RE(-) artefacts.

Table 7: Recommended ownership of QA rectification actions

9 Conclusion

The QA audit concludes that the Requirements Engineering process used in the Abyss Navigator Suite project demonstrates a generally structured approach to requirement development and validation.

However, several improvements are required in documentation discipline, retained process evidence, and governance clarity in order to strengthen confidence that the process strictly complied with the expected conduct of Analysing, Specifying, and Validating.

The QA auditor does not directly modify the audited process records. Instead, the findings and recommended process rectifications in this report should be used to improve the discipline, transparency, and auditability of the RE(-) process.

Overall, the RE(-) process is assessed as **conditionally acceptable subject to rectification of the identified process non-compliance issues**.

10 Sign-off

10.1 QA Audit Authentication

Name	Role	Authentication
Zong Han	QA Auditor	Submitted by named auditor to Customer C-staff

Table 8: QA audit sign-off

11 Addenda

11.1 Repository Information

PROJECT REPOSITORY https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113 <i>QA report, SRS, models, and supporting process artefacts maintained under version control</i>
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11.2 Glossary

Term	Definition
QA	Quality Assurance audit of the process used to produce the requirements work product.
RE(-) process	The vendor-side conduct of Requirements Engineering methods relevant to producing and validating the SRS.
Process non-compliance	A process weakness or deficiency indicating incomplete adherence to expected method conduct or supporting evidence.
Regulated-revision	Controlled revision of the Project Specification into improved requirements content for the SRS.
Traceability governance	The process discipline used to establish, maintain, and justify traceability links across requirements artefacts.

Table 9: QA glossary addendum

References

- Triton Deepwater, *Project Specification: The Abyss Navigator Suite*, ICT2113, 2026.
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